

In linear programming, what does the term "objective function" refer to?

- ☐ A constraint that must not be violated
- ☒ A function that maximizes or minimizes a certain value
- ☐ A model that predicts future values
- ☐ A graph showing tradeoff analysis

Correct answer

Question 3

2 / 2 pts

Which of these Python libraries is most commonly used for data manipulation?

- ☐ NumPy
- ☒ pandas
- ☐ matplotlib
- ☐ TensorFlow

Correct answer

Question 4

2 / 2 pts

What is the primary purpose of using a decision tree in business analytics?

- ☐ To generate random forecasts
- ☒ To analyze and visualize decision paths and outcomes
- ☐ To manipulate datasets in SQL
- ☐ To create dashboards for storytelling

Question 5

1 / 2 pts

Fill in the blank using the concept discussed in class - not your personal/professional opinion):

The concept to focus the audience's attention on a visual in milliseconds is called the preattentive attribute. It is all about

understanding how our visual perception works and knowing the first thing that your audience will pay attention to.

Answer 1:

- ☒ preattentive attributes
- ☐ preattentive attribute
- ☐ Preattentive Attributes
- ☐ pre-attentive attributes
- ☐ pre-attentive attribute
- ☐ Pre-Attentive Attributes

- Pre-Attentive Attribute
- Pre-attentive Attributes
- Pre-attentive Attribute
- Preattentive attributes
- Preattentive attribute

Answer 2:

- ☒ visual perception
- brain
- brains

Correct answer

Question 6

2 / 2 pts

Attributes like , ,
 and are preattentive

attributes. These are the **general** categories of preattentive attributes (not the subcategories). You must use the terms we used in this class. (Your answer should be based on class materials/discussion - not your personal/professional opinion)

Answer 1:

- ☒ color
- size
- orientation
- texture

Answer 2:

- ☒ size
- color
- orientation
- texture

Answer 3:

- ☒ orientation
- color
- size
- texture

Answer 4:

- ☒ texture
- color
- size
- orientation

Correct answer

Question 7

2 / 2 pts

Which type of charts to use when showing progression over time of

one or more groups? (Best answer based on class materials/discussion - not your personal/professional opinion)

- ☐ Bar
- ☒ Line
- ☐ Scatter
- ☐ Pie
- ☐ Histogram

Question 8

8 / 10 pts

You are the production manager of your company that produces two types of woodworking products: Product A and Product B. You have a limited amount of resources available, including labor hours and raw materials, and you want to determine the optimal production levels for each product to maximize your profits.

Let's say that you have a total of 500 labor hours available and 800 pounds of raw materials available. Producing one unit of Product A requires 2 labor hours and 3 pounds of raw materials while producing one unit of Product B requires 4 labor hours and 2 pounds of raw materials.

The production levels are non-negative.

Let's say that Product A sells for \$10 per unit and has a production cost of \$6 per unit, while Product B sells for \$12 per unit and has a production cost of \$8 per unit. Your objective is to maximize profits.

Please write your Python program with the PuLP Module to calculate the optimal solution.

Your Answer:

```
from pulp import LpMaximize, LpProblem, LpVariable, value

# model to maximize profit
model = LpProblem("woodworking_prod_optimization", LpMaximize)

# decision vars: how many of A and B to make (can't be negative)
Product_A = LpVariable("prod_A", lowBound=0)
Product_B = LpVariable("prod_B", lowBound=0)

# objective: max total profit ($4 each of A and B)
model += 4 * Product_A + 4 * Product_B, "profit"

# add constraints
model += (2 * Product_A + 4 * Product_B <= 500, "labor_hours") #
<= 500 labor hours
model += (3 * Product_A + 2 * Product_B <= 800, "raw_materials")
# <= 800 materials

# solve for best combo of A and B
model.solve()

# show how much of each to make
for var in model.variables():
    print(f"{var.name} = {var.varValue}")

print(f"max profit = ${value(model.objective)}")
```

Your model is set up correctly, but the submitted code has an indentation issue that would make it difficult to run as written. It would also be better to clearly summarize the final solution in words, including the optimal production quantities and maximum profit.

Correct answer

Question 9

2 / 2 pts

In SQL, the **GROUP BY** clause is used to aggregate data based on specific columns.

☒ True

☐ False

Correct answer

Question 10

2 / 2 pts

The newsvendor problem assumes that all demand variability can be predicted accurately.

☐ True

☒ False

Correct answer

Question 11

2 / 2 pts

Linear programming is used to find the best possible solution to a problem within given constraints.

☒ True

☐ False

Correct answer

Question 12

3 / 3 pts

In SQL, the _____ command is used to retrieve specific data from a database table.

☒ SELECT

SELECT

select

Correct answer

Question 13

3 / 3 pts

In storytelling with data, a bar chart is ideal for comparing _____ data.

☒ categorical

categorical

Wrong answer

Question 14

0 / 3 pts

The newsvendor problem involves finding the optimal inventory level to balance _____ and demand variability.



the cost of understocking

cost

Correct answer

Question 15

3 / 3 pts

A company has a net income of \$80,000, and total equity of \$400,000. The return on equity (ROE) is _____%.



20

20 (with margin: 0)

Correct answer

Question 16

3 / 3 pts

A company has current assets of \$150,000 and current liabilities of \$75,000. The current ratio is ____

NOTE: numerical answer



2

2 (with margin: 0)

Correct answer

Question 17

3 / 3 pts

The operating margin of a company with operating income of \$60,000 and total revenue of \$300,000 is _____%.

NOTE: numerical answer



20

20 (with margin: 0)

Correct answer

Question 18

4 / 4 pts

An investment offers two outcomes:

- **Outcome 1:** Profit = \$50,000 with a probability of 70%.
- **Outcome 2:** Loss = \$10,000 with a probability of 30%.

What is the expected value of this investment?

NOTE: numerical answer (enter numbers only)



32 000

32,000 (with margin: 0)

Correct answer

Question 19

4 / 4 pts

A store is considering two marketing campaigns:

- **Campaign A:** \$20,000 cost, \$50,000 profit (70%), no profit (30%).
- **Campaign B:** \$15,000 cost, \$30,000 profit (90%), no profit (10%).

Which campaign has a higher net EV?

☒ Campaign A

☐ Campaign B

Correct answer

Question 20

4 / 4 pts

A company has two options for a new project:

- **Option A:** Probability of success = 60%, profit = \$100,000; Probability of failure = 40%, loss = \$30,000.

What is the expected value (EV) of Option A?

NOTE: numerical answer (enter numbers only)

☒ 48,000

48,000 (with margin: 0)

Correct answer

Question 21

3 / 3 pts

A retailer sells a product for \$70 with a cost of \$42 per unit. Unsold units have a salvage value of \$22. The demand is normally distributed with a mean of 470 and a standard deviation of 85.

What is the critical ratio (CR)?

NOTE: numerical answer (enter numbers only)

☒ 0.5833

0.583 (with margin: 0.005)

Question 22

30 / 30 pts

Final Exam Problem: FestiveCo Holiday Gift Delivery Optimization with Tight Deadlines

Scenario:

FestiveCo, a holiday gift company, needs to optimize its distribution network to deliver gifts to three regions: East, Central, and West. Due to varying customer expectations, each region has strict

delivery deadlines that cannot be exceeded. Additionally, FestiveCo operates three fulfillment centers with limited stock and uses multiple transportation methods with varying costs and delivery times.

Your task is to minimize the total transportation costs while meeting customer demands, adhering to stock limits, and ensuring that all deliveries are completed within the required deadlines.

Input Data:

1. Fulfillment Centers:

- Boston: 500 units
- Denver: 400 units
- Los Angeles: 300 units

2. Customer Regions and Deadlines:

- East:
 - Demand: 450 units
 - Delivery Deadline: 5 days
- Central:
 - Demand: 350 units
 - Delivery Deadline: 10 days
- West:
 - Demand: 300 units
 - Delivery Deadline: 15 days

3. Transportation Costs and Delivery Times:

- Boston to East:
 - Ground: \$10 per unit, 3 days
 - Air: \$25 per unit, 1 day
- Denver to Central:
 - Ground: \$15 per unit, 5 days
 - Air: \$35 per unit, 2 days
- Los Angeles to West:
 - Ground: \$20 per unit, 7 days
 - Air: \$50 per unit, 3 days
- Boston to Central:
 - Ground: \$18 per unit, 6 days
 - Air: \$40 per unit, 2 days
- Denver to East:
 - Ground: \$22 per unit, 8 days (Does not meet East's deadline)
 - Air: \$45 per unit, 3 days
- Los Angeles to East:
 - Ground: \$30 per unit, 10 days (Does not meet East's deadline)
 - Air: \$60 per unit, 4 days

Constraints:

1. Stock Limits:

- Boston can ship at most 500 units.
- Denver can ship at most 400 units.
- Los Angeles can ship at most 300 units.

2. Demand Satisfaction:

- East must receive exactly 450 units.
- Central must receive exactly 350 units.
- West must receive exactly 300 units.

3. Regional Delivery Deadlines:

- Shipments to East must arrive within **5 days**.
- Shipments to Central must arrive within **10 days**.
- Shipments to West must arrive within **15 days**.

Objective:

Minimize the total transportation costs while satisfying all constraints.

Questions:

1. Formulate the Problem:

- Define the decision variables.
- Write the objective function to minimize total transportation costs.
- Specify the constraints for stock limits, demand satisfaction, and delivery deadlines.

2. Solve the Problem in Python:

- Use the PuLP library to implement your linear programming model.
- Provide the Python code and the optimal solution.

3. Analyze the Solution:

- What is the optimal total cost?
- How many units should be shipped on each route, and which transportation methods are used?

4. What-If Analysis:

- Suppose the deadline for the East region is extended to **10 days**. How does this affect the optimal solution and total cost?
- Suppose the Central region's demand increases by **50 units**. Can the fulfillment centers meet the demand under the current setup?

Hints for Students:

1. Consider the delivery deadline for each region when selecting transportation methods.
2. Routes that do not meet a region's deadline should not appear in the solution (e.g., ground shipping from Denver to East).
3. Ensure your code is modular and well-documented so that you can adjust parameters for the what-if analysis.
4. Use text box provided below to enter your answers, code, and analysis.

Your Answer:

1. Formulating the Problem

At a high level, we're using a standard transportation network optimization model. The main formula is to Minimize the Objective Function =

$\text{Sum}(\text{Cost}_{ijm} * x_{ijm})$ across all valid routes, subject to the constraints that

$\text{Sum}(\text{Units shipped from Origin } i) \leq \text{Supply at } i$, and

$\text{Sum}(\text{Units shipped to Destination } j) = \text{Demand at } j$

Before listing variables, I'm completely excluding any routes that do not meet the strict delivery deadlines (specifically, Denver to East Ground and LA to East Ground) since they're invalid options

Decision Variables:

Let x_{ijm} represent the number of units shipped from origin i to destination j via transportation method m

Valid variables: x_{BEG} , x_{BEA} , x_{BCG} , x_{BCA} , x_{DCG} , x_{DCA} , x_{DEA} , x_{LWG} , x_{LWA} , x_{LEA}

Objective Function:

Minimize Total Cost (Z) = $10(x_{\text{BEG}}) + 25(x_{\text{BEA}}) + 18(x_{\text{BCG}}) + 40(x_{\text{BCA}}) + 15(x_{\text{DCG}}) + 35(x_{\text{DCA}}) + 45(x_{\text{DEA}}) + 20(x_{\text{LWG}}) + 50(x_{\text{LWA}}) + 60(x_{\text{LEA}})$

Constraints:

1. Supply Limits (cannot ship more than available stock):

Boston: $\text{Sum}(x_{\text{B}...}) = x_{\text{BEG}} + x_{\text{BEA}} + x_{\text{BCG}} + x_{\text{BCA}} \leq 500$

Denver: $\text{Sum}(x_{\text{D}...}) = x_{\text{DCG}} + x_{\text{DCA}} + x_{\text{DEA}} \leq 400$

Los Angeles: $\text{Sum}(x_{\text{L}...}) = x_{\text{LWG}} + x_{\text{LWA}} + x_{\text{LEA}} \leq 300$

2. Demand Satisfaction (must meet exact regional demand):

East: $\text{Sum}(x_{\text{...E}...}) = x_{\text{BEG}} + x_{\text{BEA}} + x_{\text{DEA}} + x_{\text{LEA}} = 450$

Central: $\text{Sum}(x_{\text{...C}...}) = x_{\text{BCG}} + x_{\text{BCA}} + x_{\text{DCG}} + x_{\text{DCA}} = 350$

West: $\text{Sum}(x_{\dots W\dots}) = x_{\text{LWG}} + x_{\text{LWA}} = 300$

3. Non-negativity:

All $x_{ijm} \geq 0$

2. Code for Calculations

```
from pulp import LpProblem, LpMinimize, LpVariable, lpSum, LpStatus

model = LpProblem("FestiveCo_delivery_plan", LpMinimize)

# valid shipping routes: (origin, destination, shipping method)
# note: ground routes to east/central from denver and la are left out due to delivery deadlines
routes = [
    ("boston", "east", "ground"),
    ("boston", "east", "air"),
    ("boston", "central", "ground"),
    ("boston", "central", "air"),
    ("denver", "central", "ground"),
    ("denver", "central", "air"),
    ("denver", "east", "air"),
    ("la", "west", "ground"),
    ("la", "west", "air"),
    ("la", "east", "air"),
]

# cost per unit for each route
costs = {
    ("boston", "east", "ground"): 10,
    ("boston", "east", "air"): 25,
    ("boston", "central", "ground"): 18,
    ("boston", "central", "air"): 40,
    ("denver", "central", "ground"): 15,
    ("denver", "central", "air"): 35,
    ("denver", "east", "air"): 45,
    ("la", "west", "ground"): 20,
    ("la", "west", "air"): 50,
    ("la", "east", "air"): 60,
}

# decision vars: how much to send on each route (continuous  $\geq 0$ )
x = LpVariable.dicts("route", routes, lowBound=0, cat="Continuous")

# minimize total shipping cost
model += lpSum([x[r] * costs[r] for r in routes]), "total_delivery_cost"

# constraints:

# 1. stock limits
model += lpSum([x[r] for r in routes if r[0] == "boston"]) <= 500, "stock_boston"
model += lpSum([x[r] for r in routes if r[0] == "denver"]) <= 400, "stock_denver"
model += lpSum([x[r] for r in routes if r[0] == "la"]) <= 300, "stock_la"

# 2. demand satisfaction
model += lpSum([x[r] for r in routes if r[1] == "east"]) == 450, "demand_east"
model += lpSum([x[r] for r in routes if r[1] == "central"]) == 350, "demand_central"
model += lpSum([x[r] for r in routes if r[1] == "west"]) == 300, "demand_west"

# 3. regional delivery deadlines handled by omitted routes
# already handled this by omitting them above in 'routes'

# run opt
model.solve()
```

```
# show solution
print(f"solver status: {LpStatus[model.status].lower()}")
print(f"minimum total shipping cost:
${model.objective.value():.2f}")
print("recommended shipping plan:")
for r in routes:
    if x[r].varValue and x[r].varValue > 0:
        print(f"\tship {x[r].varValue} units from {r[0]} to {r[1]} via
{r[2]}")
```

3. Analysis

Optimal Total Cost: **\$15,750**

Breakdown of the most optimal routing:

1. Ship 300 units from LA to West via Ground. (LA is the only origin that services the West, and Ground is the cheapest valid method to meet the 15-day deadline)
2. Ship 450 units from Boston to East via Ground. (This uses Boston's cheapest method to fully satisfy the East's demand within the 5-day deadline)
3. Ship 350 units from Denver to Central via Ground. (This satisfies Central's demand using the cheapest remaining valid method).

* All other routes are strictly 0 units.

4. What-If Analysis

Scenario A: East deadline extended to 10 days

This change has zero effect on the optimal solution or the total cost. Extending the deadline simply unlocks the Denver-to-East Ground (\$22/u) and LA-to-East Ground (\$30/u) routes.

as valid options in the model. However, the current Boston-to-East Ground route is still much cheaper at \$10/unit. Because Boston has a total supply of 500 units, which is more than enough to cover the entire East demand of 450 units. The model naturally sticks with the cheaper route so our total cost will still stick at \$15,750.

Scenario B: Central demand increases by 50 units

Yes, the fulfillment centers can easily meet this demand under the current setup. The new total network demand is 1,150 units, while total network supply is 1,200 units.

Looking at our optimal solution, Denver is only shipping 350 of its 400 available units to Central. Denver has 50 units of unused supply. To meet this new demand, Denver can just ship those remaining 50 to Central through ground. This satisfies our new demand constraint without violating any of the constraints we have above.

Correct answer

Question 23

2 / 2 pts

The Bruin Company's final results are as follows for 2022:

- Working Capital = \$500,000
- Total Assets = \$1,500,000
- Retained Earnings = \$300,000
- EBIT = \$200,000
- Market Value of Equity = \$1,000,000
- Book Value of Total Liabilities = \$500,000
- Sales = \$3,000,000

Using Altman's Z-Score, calculate the Altman Z-score for the Bruin Company using its 2022 financial data.

The answer should be rounded up to two decimal places. For example: 1.23



4.32

4.32 (with margin: 0)

Question 24

2 / 2 pts

What common modeling problem occurred in the Domino's, Boeing, and Amazon cases?

Your Answer:

The common problem was that the objective function was incomplete. All 3 companies over optimized for their speed and short-term cost reduction rather than factoring in safety risks, human costs, and long term liabilities.

Correct answer**Question 25**

3 / 3 pts

A retailer sells a product for **\$50** with a cost of **\$30 per unit**. Unsold units have a salvage value of **\$10**.

Demand is normally distributed with:

- mean $\mu = 500$
- standard deviation $\sigma = 100$

From the critical fractile calculation, we obtain:

$z = 0.43$

What is the optimal order quantity Q^* ?

NOTE: numerical answer only (enter numbers only)



543 (with margin: 0)

Quiz Score: **94** out of 100

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